

CALCULUS & ALGEBRA

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1. DERIVATIVE RULES (THE BASICS)

- **Power Rule:** $\frac{d}{dx}(x^n) = nx^{n-1}$
- **Constant Multiple:** $\frac{d}{dx}(cf(x)) = cf'(x)$
- **Sum Rule:** $\frac{d}{dx}(f \pm g) = f' \pm g'$
- **Product Rule:** $\frac{d}{dx}(uv) = u'v + uv'$
- **Quotient Rule:** $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$
- **Chain Rule:** $\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$

2. COMMON DERIVATIVES

Memorize these. They appear everywhere.

- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(a^x) = a^x \ln(a)$
- $\frac{d}{dx}(\ln x) = \frac{1}{x}$
- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$

3. ML ACTIVATION DERIVATIVES

Crucial for Backpropagation questions!

Sigmoid Function $\sigma(x)$:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$

Tanh Function $\tanh(x)$:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\tanh'(x) = 1 - \tanh^2(x)$$

ReLU (Rectified Linear Unit):

$$f(x) = \max(0, x)$$

$$f'(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$$

4. INTEGRATION FORMULAS

Used in Probability (finding Area under PDF).

Basic Rules:

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
- $\int \frac{1}{x} dx = \ln|x| + C$
- $\int e^x dx = e^x + C$

Definite Integral (Area):

$$\int_a^b f(x) dx = F(b) - F(a)$$

5. MATRIX CALCULUS IDENTITIES

The "Secret Weapon" for MFML. Assume vectors $\mathbf{x}$, $\mathbf{a}$, $\mathbf{b}$ and matrix $\mathbf{A}$.

Scalar-by-Vector Gradients:

$$\nabla_{\mathbf{x}}(\mathbf{a}^T \mathbf{x}) = \mathbf{a}$$

$$\nabla_{\mathbf{x}}(\mathbf{x}^T \mathbf{x}) = 2\mathbf{x}$$

$$\nabla_{\mathbf{x}}(\mathbf{x}^T \mathbf{A} \mathbf{x}) = (\mathbf{A} + \mathbf{A}^T)\mathbf{x}$$

Note: If $\mathbf{A}$ is Symmetric, this becomes $2\mathbf{A}\mathbf{x}$.

6. LIMITS (L'HOPITAL'S RULE)

If $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ results in $\frac{0}{0}$ or $\frac{\infty}{\infty}$:

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

7. IMPORTANT ALGEBRA PROPERTIES

Logarithms (Log-Likelihoods):

- $\ln(ab) = \ln a + \ln b$
- $\ln(a/b) = \ln a - \ln b$
- $\ln(a^b) = b \ln a$

8. OPTIMIZATION QUICK-GUIDE

Gradient Descent Update:

$$w_{new} = w_{old} - \alpha \nabla J(w)$$

Finding Min/Max (Univariate):

1. Find $f'(x) = 0$ (Critical Points).
2. Check $f''(x)$:
 - $f''(x) > 0 \implies$ Local Min (Convex).
 - $f''(x) < 0 \implies$ Local Max (Concave).

9. Exam Strategy Tips

1. **Chain Rule Trap:** Inner derivatives!

$$\frac{d}{dx}(y - \hat{y})^2 = 2(y - \hat{y}) \cdot (-1)$$

2. **Matrix Dimensions:** $(m \times n) \times (n \times p) = (m \times p)$
3. **Product Rule:** Don't just derivate terms separately!

WORKED EXAMPLES: CALCULUS

Solutions for Derivatives, Chain Rule, and Integration

EXAMPLES: DERIVATIVE RULES

EXAMPLE: Product Rule

Q: Differentiate $f(x) = x^2 \sin(x)$.

Solution: Apply $u'v + uv'$.

- $u = x^2 \implies u' = 2x$

- $v = \sin x \implies v' = \cos x$

$$f'(x) = (2x)(\sin x) + (x^2)(\cos x)$$

EXAMPLE: Quotient Rule

Q: Differentiate $f(x) = \frac{e^x}{x}$.

Solution: Apply $\frac{u'v - uv'}{v^2}$.

$$f'(x) = \frac{(e^x)(x) - (e^x)(1)}{x^2} = \frac{e^x(x - 1)}{x^2}$$

EXAMPLE: Chain Rule (Important!)

Q: Differentiate $f(x) = (3x^2 + 5)^4$.

Solution: Outer function is $(\dots)^4$, Inner is $3x^2 + 5$.

$$f'(x) = 4(3x^2 + 5)^3 \cdot \frac{d}{dx}(3x^2 + 5)$$

$$f'(x) = 4(3x^2 + 5)^3 \cdot (6x) = 24x(3x^2 + 5)^3$$

EXAMPLES: ACTIVATIONS

EXAMPLE: Sigmoid Calculation

Q: Given $\sigma(x) = 0.8$, find $\sigma'(x)$.

Solution: Use identity $\sigma'(x) = \sigma(x)(1 - \sigma(x))$.

$$\sigma'(x) = 0.8(1 - 0.8)$$

$$\sigma'(x) = 0.8(0.2) = 0.16$$

EXAMPLE: Tanh Calculation

Q: Given $\tanh(x) = 0.5$, find $\tanh'(x)$.

Solution: Use identity $\tanh'(x) = 1 - \tanh^2(x)$.

$$\tanh'(x) = 1 - (0.5)^2$$

$$\tanh'(x) = 1 - 0.25 = 0.75$$

EXAMPLES: INTEGRATION

EXAMPLE: Definite Integral (Area)

Q: Find area under $f(x) = 3x^2$ from 0 to 2.

Solution: Integrate first:

$$\int 3x^2 dx = x^3$$

Evaluate bounds $F(2) - F(0)$:

$$[2^3] - [0^3] = 8 - 0 = 8$$

EXAMPLE: Prob. Normalization

Q: Find C so $p(x) = Ce^{-2x}$ is a valid PDF for $x > 0$.

Solution: Total area must be 1.

$$\int_0^\infty Ce^{-2x} dx = 1$$

$$C \left[\frac{e^{-2x}}{-2} \right]_0^\infty = 1$$

$$C[0 - (-1/2)] = 1 \implies C(0.5) = 1$$

$$C = 2$$

WORKED EXAMPLES: MATRIX & OPTIMIZATION

Solutions for Matrix Gradients, Log-Likelihood, and Min/Max

EXAMPLES: MATRIX CALCULUS

EXAMPLE: Gradient of Dot Product

Q: Find $\nabla_{\mathbf{x}}(\mathbf{x}^T \mathbf{a})$.

Solution: Let $\mathbf{x} = [x_1, x_2]^T$, $\mathbf{a} = [3, 5]^T$.

$$f = 3x_1 + 5x_2$$

$$\nabla_{\mathbf{x}} f = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \mathbf{a}$$

EXAMPLE: Least Squares Derivation

Q: Find Gradient of $J = (\mathbf{A}\mathbf{x} - \mathbf{b})^T (\mathbf{A}\mathbf{x} - \mathbf{b})$.

Solution: Expand first:

$$J = \mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x} - 2\mathbf{x}^T \mathbf{A}^T \mathbf{b} + \mathbf{b}^T \mathbf{b}$$

Derive term by term:

1. $\nabla(\mathbf{x}^T (\mathbf{A}^T \mathbf{A}) \mathbf{x}) = 2\mathbf{A}^T \mathbf{A} \mathbf{x}$ (Symmetric rule)
2. $\nabla(-2\mathbf{x}^T \mathbf{A}^T \mathbf{b}) = -2\mathbf{A}^T \mathbf{b}$
3. $\nabla(\text{constant}) = 0$

$$\nabla J = 2\mathbf{A}^T \mathbf{A} \mathbf{x} - 2\mathbf{A}^T \mathbf{b}$$

EXAMPLES: ALGEBRA MLE

EXAMPLE: Log Expansion

Q: Simplify $\ln\left(\frac{e^{2x} y^3}{\sqrt{z}}\right)$.

Solution: Use log laws step-by-step: 1. Division: $\ln(e^{2x} y^3) - \ln(z^{0.5})$ 2. Product: $\ln(e^{2x}) + \ln(y^3) - \ln(z^{0.5})$ 3. Powers: $2x + 3 \ln y - 0.5 \ln z$

EXAMPLE: L'Hopital's Rule

Q: Find $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$.

Solution: Plugging in 0 gives 0/0. Derive top and bottom.

$$\lim_{x \rightarrow 0} \frac{e^x}{1} = e^0 = 1$$

EXAMPLES: OPTIMIZATION

EXAMPLE: Find Minimum (Univariate)

Q: Minimize $J(w) = w^2 - 4w + 5$.

Solution: Step 1: Find Critical Point ($J' = 0$).

$$J'(w) = 2w - 4 = 0 \implies w = 2$$

Step 2: Check Convexity (J'').

$$J''(w) = 2 \quad (> 0, \text{ so it is a Min})$$

EXAMPLE: Gradient Descent Step

Q: $w_{old} = 1, \alpha = 0.1, J'(w) = 2w - 4$. Calculate w_{new} .

Solution: Evaluate gradient at old w :

$$\nabla J(1) = 2(1) - 4 = -2$$

Update rule:

$$w_{new} = 1 - (0.1)(-2)$$

$$w_{new} = 1 + 0.2 = 1.2$$